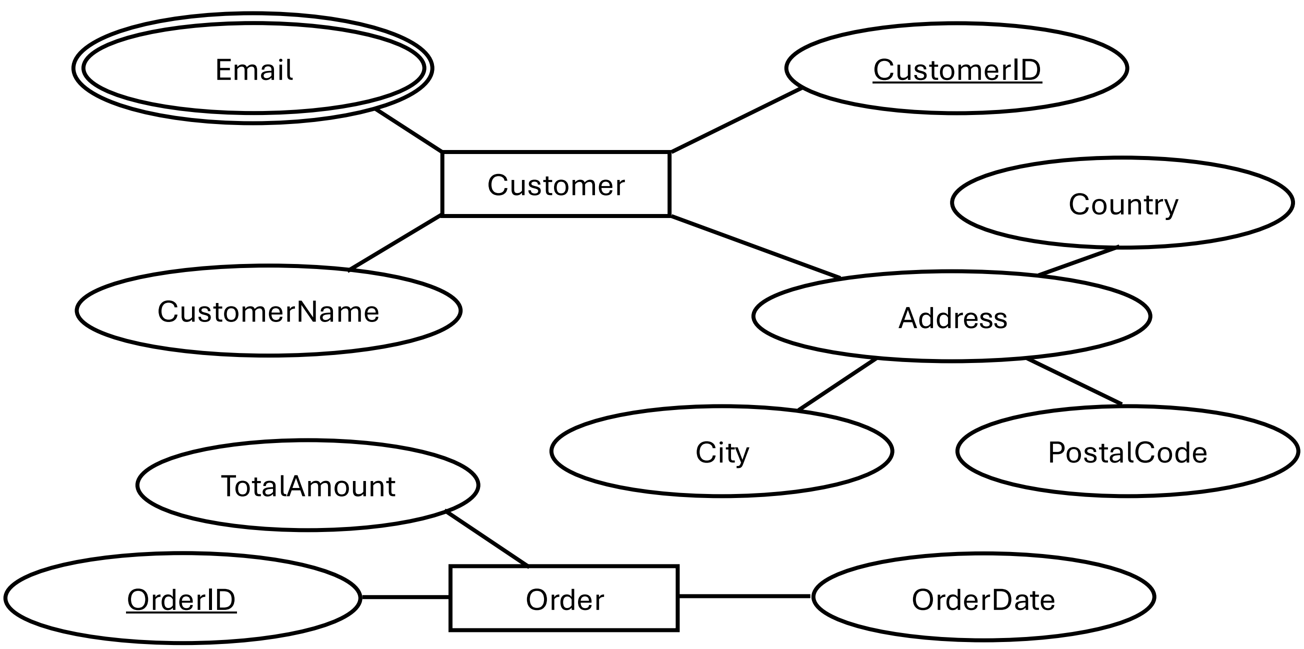
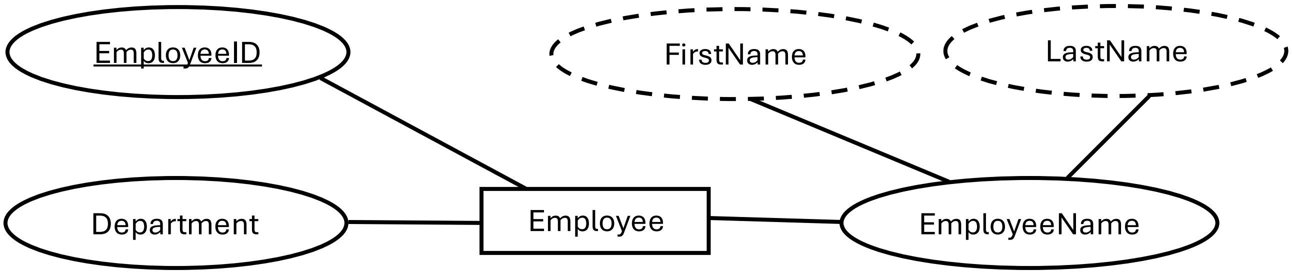
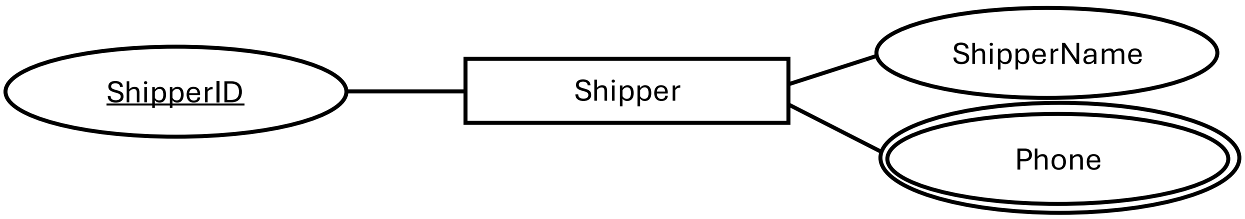
Orders Processing System – ER Diagram

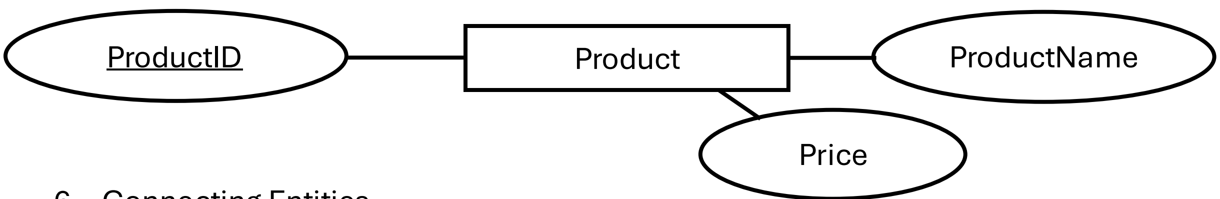
1. Entities
   - Strong Entities: Customer, Order, Employee, Shipper, Product
   - Weak Entities:
2. Attributes
   - Customer: CustomerID, CustomerName, Phone, Address, City, PostalCode, Country
   - Order: OrderID, OrderDate, TotalAmount
   - Employee: EmployeeID, FirstName, LastName, Department
   - Shipper: ShipperID, ShipperName, Phone
   - Product: ProductID, ProductName, Price
3. Relationships
   - Customer PLACES Order
   - Employee PROCESSES Order
   - Shipper SHIPS Order
   - Order CONTAINS Product
4. Drawing Entities

5. Adding Attributes

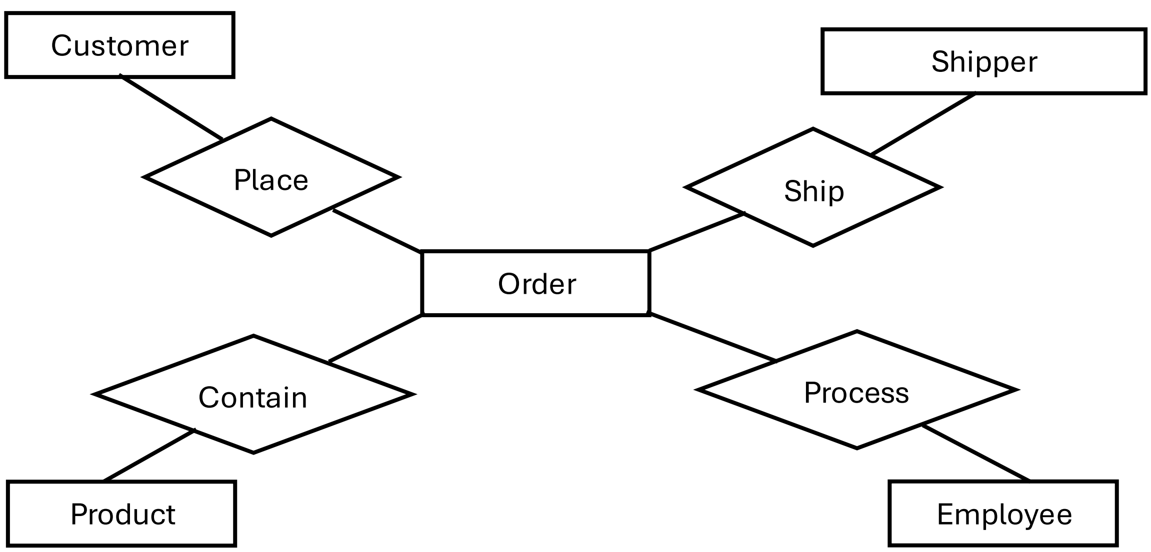




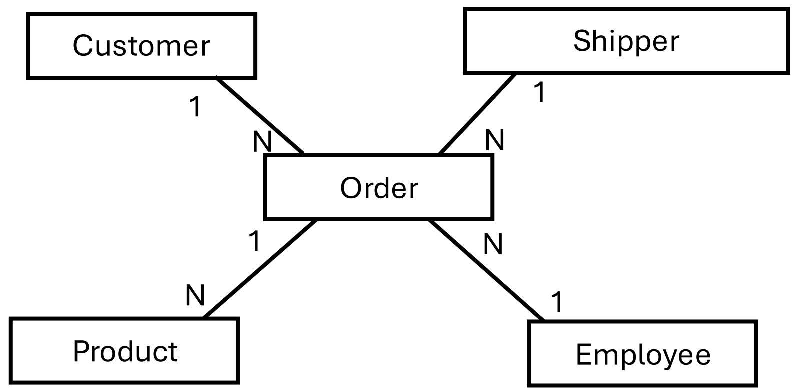




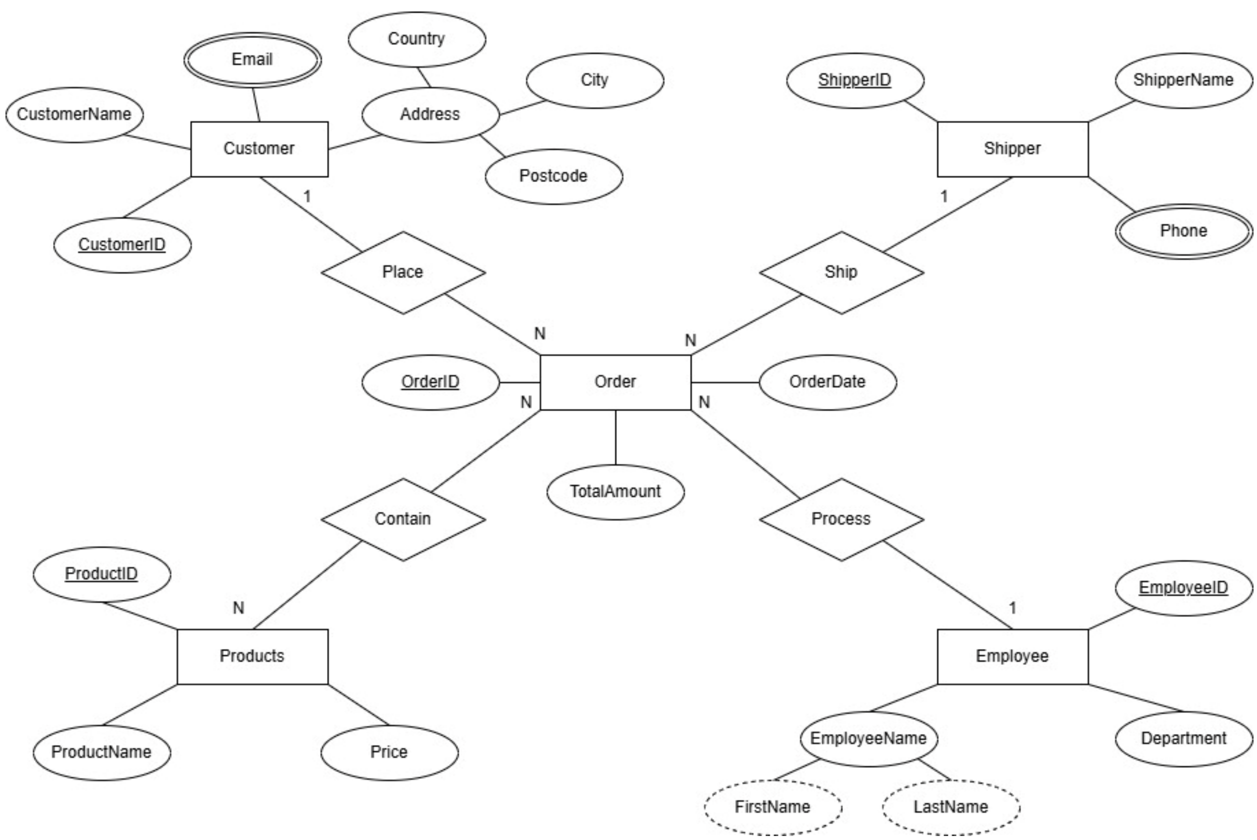
6. Connecting Entities



7. Specifying Cardinality



8. Organizing ER Diagram



<u>Order Processing System – Table Operations</u>

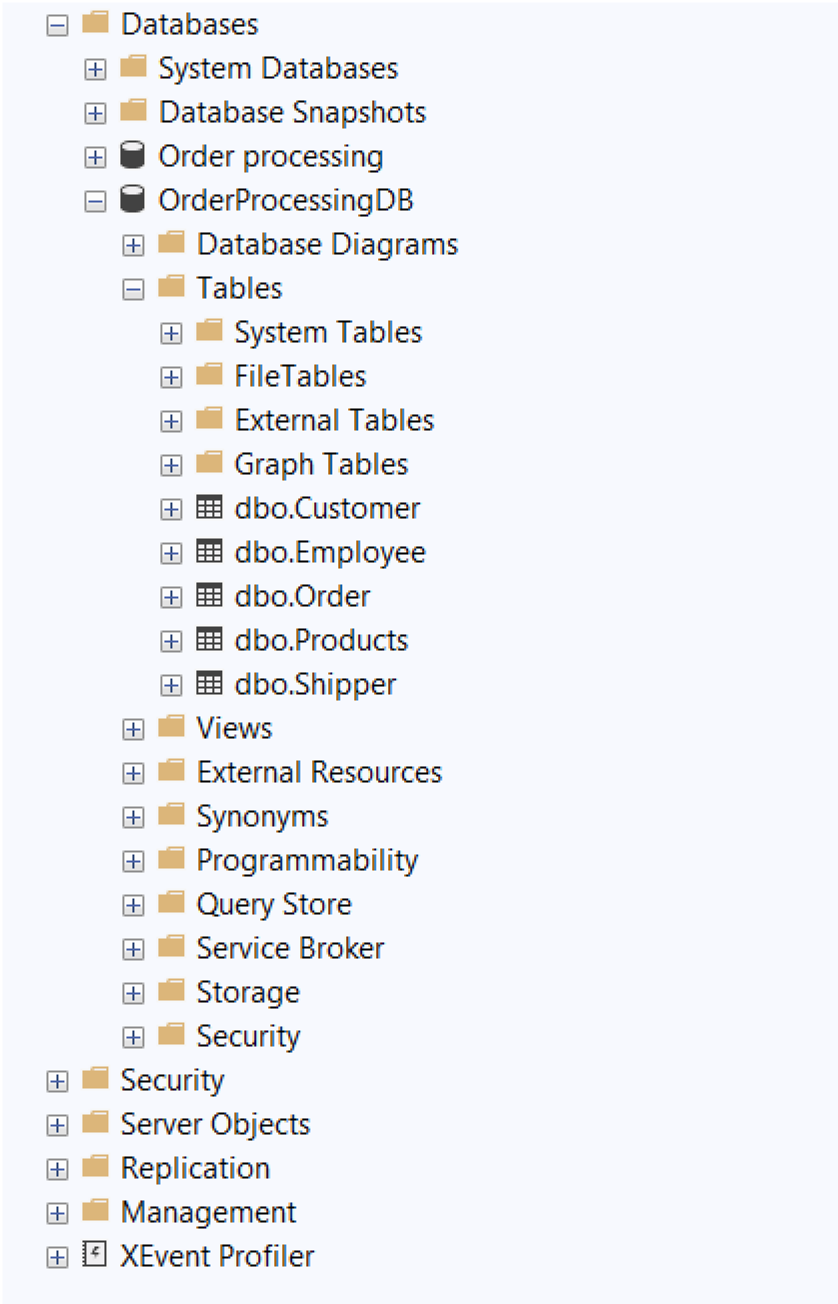
Creating Database:

```
CREATE DATABASE OrderProcessingDB;

USE OrderProcessingDB;
```

Creating Tables:

```
CREATE TABLE Customer (
   CustomerID INT PRIMARY KEY,
   CustomerName NVARCHAR(100) NOT NULL,
   Email NVARCHAR(100),
   Address NVARCHAR(255),
   City NVARCHAR(50),
   Postcode NVARCHAR(20),
   Country NVARCHAR(50)
);

CREATE TABLE Shipper (
   ShipperID INT PRIMARY KEY,
   ShipperName NVARCHAR(100) NOT NULL,
   Phone NVARCHAR(20)
);

CREATE TABLE Employee (
   EmployeeID INT PRIMARY KEY,
   FirstName NVARCHAR(50),
   LastName NVARCHAR(50),
```

```
    Department NVARCHAR(50)
);

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName NVARCHAR(100) NOT NULL,
    Price DECIMAL(10, 2) NOT NULL
);

CREATE TABLE [Order] (
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    TotalAmount DECIMAL(10, 2),
    CustomerID INT,
    ShipperID INT,
    EmployeeID INT,
    ProductID INT, -- Linking directly to Products for your 1:N requirement
    CONSTRAINT FK_Customer FOREIGN KEY (CustomerID) REFERENCES
Customer(CustomerID),
    CONSTRAINT FK_Shipper FOREIGN KEY (ShipperID) REFERENCES Shipper(ShipperID),
    CONSTRAINT FK_Employee FOREIGN KEY (EmployeeID) REFERENCES
Employee(EmployeeID),
    CONSTRAINT FK_Product FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);
```


Databases
System Databases
Database Snapshots
Order processing
OrderProcessingDB
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.Customer
dbo.Employee
dbo.Order
dbo.Products
dbo.Shipper
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security
Security
Server Objects
Replication
Management
XEvent Profiler

Inserting Data:

INSERT INTO Customer (CustomerID, CustomerName, Email, Address, City, Postcode, Country) VALUES

(1, 'Alice Smith', 'alice@email.com', '123 Maple St', 'New York', '10001', 'USA'),

(2, 'Bob Jones', 'bob@email.com', '456 Oak Ave', 'London', 'W1D', 'UK'),

(3, 'Charlie Brown', 'charlie@email.com', '789 Pine Rd', 'Paris', '75001', 'France'),

(4, 'Diana Prince', 'diana@email.com', '101 Elm St', 'Berlin', '10115', 'Germany'),

(5, 'Evan Wright', 'evan@email.com', '202 Cedar Ln', 'Toronto', 'M5V', 'Canada'),

(6, 'Fiona Green', 'fiona@email.com', '303 Birch Blvd', 'Sydney', '2000', 'Australia'),

(7, 'George Hall', 'george@email.com', '404 Spruce Way', 'Tokyo', '100-00', 'Japan'),

(8, 'Hannah Lee', 'hannah@email.com', '505 Fir Dr', 'Seoul', '045', 'South Korea'),

(9, 'Ian Scott', 'ian@email.com', '606 Ash Ct', 'Dublin', 'D01', 'Ireland'),

(10, 'Jane Doe', 'jane@email.com', '707 Willow Pl', 'Rome', '00100', 'Italy'),

(11, 'Kyle Reese', 'kyle@email.com', '808 Aspen St', 'Madrid', '28001', 'Spain'),

(12, 'Laura Croft', 'laura@email.com', '909 Poplar Ln', 'Lisbon', '1000', 'Portugal'),

(13, 'Mike Ross', 'mike@email.com', '111 Redwood Dr', 'New York', '10002', 'USA'),

(14, 'Nancy Drew', 'nancy@email.com', '222 Cypress Rd', 'Chicago', '60601', 'USA'),

(15, 'Oscar Wilde', 'oscar@email.com', '333 Magnolia St', 'Dublin', 'D02', 'Ireland'),

(16, 'Paul Rudd', 'paul@email.com', '444 Palm Ave', 'London', 'SW1', 'UK'),

(17, 'Quinn Fabray', 'quinn@email.com', '555 Olive Way', 'Paris', '75002', 'France'),

(18, 'Rachel Green', 'rachel@email.com', '666 Cherry Ln', 'New York', '10003', 'USA'),

(19, 'Steve Rogers', 'steve@email.com', '777 Walnut St', 'Brooklyn', '11201', 'USA'),

(20, 'Tony Stark', 'tony@email.com', '888 Malibu Pt', 'Malibu', '90265', 'USA');

SELECT * FROM Customer;

	CustomerID	CustomerName	Email	Address	City	Postcode	Country
1	1	Alice Smith	alice@email.com	123 Maple St	New York	10001	USA
2	2	Bob Jones	bob@email.com	456 Oak Ave	London	W1D	UK
3	3	Charlie Brown	charlie@email.com	789 Pine Rd	Paris	75001	France
4	4	Diana Prince	diana@email.com	101 Elm St	Berlin	10115	Germany
5	5	Evan Wright	evan@email.com	202 Cedar Ln	Toronto	M5V	Canada
6	6	Fiona Green	fiona@email.com	303 Birch Blvd	Sydney	2000	Australia
7	7	George Hall	george@email.com	404 Spruce Way	Tokyo	100-00	Japan
8	8	Hannah Lee	hannah@email.com	505 Fir Dr	Seoul	045	South Korea
9	9	Ian Scott	ian@email.com	606 Ash Ct	Dublin	D01	Ireland
10	10	Jane Doe	jane@email.com	707 Willow Pl	Rome	00100	Italy
11	11	Kyle Reese	kyle@email.com	808 Aspen St	Madrid	28001	Spain
12	12	Laura Croft	laura@email.com	909 Poplar Ln	Lisbon	1000	Portugal
13	13	Mike Ross	mike@email.com	111 Redwood Dr	New York	10002	USA
14	14	Nancy Drew	nancy@email.com	222 Cypress Rd	Chicago	60601	USA
15	15	Oscar Wilde	oscar@email.com	333 Magnolia St	Dublin	D02	Ireland
16	16	Paul Rudd	paul@email.com	444 Palm Ave	London	SW1	UK
17	17	Quinn Fabray	quinn@email.com	555 Olive Way	Paris	75002	France
18	18	Rachel Green	rachel@email.com	666 Cherry Ln	New York	10003	USA
19	19	Steve Rogers	steve@email.com	777 Walnut St	Brooklyn	11201	USA
20	20	Tony Stark	tony@email.com	888 Malibu Pt	Malibu	90265	USA

```sql
INSERT INTO Employee (EmployeeID, FirstName, LastName, Department) VALUES
(101, 'John', 'Doe', 'Sales'),
(102, 'Jane', 'Smith', 'IT'),
(103, 'Michael', 'Johnson', 'Sales'),
(104, 'Emily', 'Davis', 'HR'),
(105, 'Chris', 'Brown', 'IT'),
(106, 'Sarah', 'Wilson', 'Marketing'),
(107, 'David', 'Taylor', 'Sales'),
(108, 'Laura', 'Moore', 'Finance'),
(109, 'Robert', 'Anderson', 'IT'),
(110, 'Jennifer', 'Thomas', 'HR'),
(111, 'William', 'Jackson', 'Sales'),
(112, 'Elizabeth', 'White', 'Marketing'),
(113, 'James', 'Harris', 'Finance'),
(114, 'Patricia', 'Martin', 'IT'),
(115, 'Linda', 'Thompson', 'Sales'),
(116, 'Barbara', 'Garcia', 'HR'),
(117, 'Richard', 'Martinez', 'Marketing'),
(118, 'Susan', 'Robinson', 'Finance'),
(119, 'Joseph', 'Clark', 'IT'),
(120, 'Thomas', 'Rodriguez', 'Sales');
```

SELECT * FROM Employee;

	EmployeeID	FirstName	LastName	Department
1	101	John	Doe	Sales
2	102	Jane	Smith	IT
3	103	Michael	Johnson	Sales
4	104	Emily	Davis	HR
5	105	Chris	Brown	IT
6	106	Sarah	Wilson	Marketing
7	107	David	Taylor	Sales
8	108	Laura	Moore	Finance
9	109	Robert	Anderson	IT
10	110	Jennifer	Thomas	HR
11	111	William	Jackson	Sales
12	112	Elizabeth	White	Marketing
13	113	James	Harris	Finance
14	114	Patricia	Martin	IT
15	115	Linda	Thompson	Sales
16	116	Barbara	Garcia	HR
17	117	Richard	Martinez	Marketing
18	118	Susan	Robinson	Finance
19	119	Joseph	Clark	IT
20	120	Thomas	Rodriguez	Sales

```sql
INSERT INTO Shipper (ShipperID, ShipperName, Phone) VALUES
(1, 'Speedy Express', '555-1000'),
(2, 'United Package', '555-1001'),
(3, 'Federal Shipping', '555-1002'),
(4, 'Global Freight', '555-1003'),
(5, 'Quick Ship', '555-1004'),
(6, 'Prime Delivery', '555-1005'),
(7, 'Fast Track', '555-1006'),
(8, 'Secure Move', '555-1007'),
(9, 'Direct Cargo', '555-1008'),
(10, 'Air Ways', '555-1009'),
(11, 'Sea Transport', '555-1010'),
(12, 'Road Runners', '555-1011'),
(13, 'City Couriers', '555-1012'),
(14, 'Nationwide', '555-1013'),
(15, 'World Wide', '555-1014'),
(16, 'Eco Ship', '555-1015'),
(17, 'Night Owl', '555-1016'),
(18, 'Day Break', '555-1017'),
(19, 'Northern Star', '555-1018'),
(20, 'Southern Cross', '555-1019');
```

SELECT * FROM Shipper;

	ShipperID	ShipperName	Phone
1	1	Speedy Express	555-1000
2	2	United Package	555-1001
3	3	Federal Shipping	555-1002
4	4	Global Freight	555-1003
5	5	Quick Ship	555-1004
6	6	Prime Delivery	555-1005
7	7	Fast Track	555-1006
8	8	Secure Move	555-1007
9	9	Direct Cargo	555-1008
10	10	Air Ways	555-1009
11	11	Sea Transport	555-1010
12	12	Road Runners	555-1011
13	13	City Couriers	555-1012
14	14	Nationwide	555-1013
15	15	World Wide	555-1014
16	16	Eco Ship	555-1015
17	17	Night Owl	555-1016
18	18	Day Break	555-1017
19	19	Northern Star	555-1018
20	20	Southern Cross	555-1019

```sql
INSERT INTO Products (ProductID, ProductName, Price) VALUES
(1, 'Laptop', 1200.00),
(2, 'Mouse', 25.00),
(3, 'Keyboard', 45.00),
(4, 'Monitor', 300.00),
(5, 'Chair', 150.00),
(6, 'Desk', 250.00),
(7, 'Lamp', 40.00),
(8, 'Pen', 1.50),
(9, 'Notebook', 3.00),
(10, 'Stapler', 5.00),
(11, 'Phone', 800.00),
(12, 'Tablet', 600.00),
(13, 'Charger', 20.00),
(14, 'Headphones', 100.00),
(15, 'Sofa', 800.00),
(16, 'Table', 400.00),
(17, 'Book', 15.00),
(18, 'Magazine', 8.00),
(19, 'Bag', 50.00),
(20, 'Watch', 150.00);
```

SELECT * FROM Products;

	ProductID	ProductName	Price
1	1	Laptop	1200.00
2	2	Mouse	25.00
3	3	Keyboard	45.00
4	4	Monitor	300.00
5	5	Chair	150.00
6	6	Desk	250.00
7	7	Lamp	40.00
8	8	Pen	1.50
9	9	Notebook	3.00
10	10	Stapler	5.00
11	11	Phone	800.00
12	12	Tablet	600.00
13	13	Charger	20.00
14	14	Headphones	100.00
15	15	Sofa	800.00
16	16	Table	400.00
17	17	Book	15.00
18	18	Magazine	8.00
19	19	Bag	50.00
20	20	Watch	150.00

```sql
INSERT INTO [Order] (OrderID, OrderDate, TotalAmount, CustomerID, ShipperID, EmployeeID, ProductID) VALUES
(1001, '2023-01-10', 1200.00, 1, 1, 101, 1),
(1002, '2023-01-11', 25.00, 2, 2, 102, 2),
(1003, '2023-01-12', 45.00, 1, 1, 103, 3),
(1004, '2023-01-13', 300.00, 3, 3, 104, 4),
(1005, '2023-01-14', 150.00, 5, 2, 105, 5),
(1006, '2023-01-15', 1200.00, 4, 1, 101, 1),
(1007, '2023-01-16', 5.00, 2, 4, 106, 10),
(1008, '2023-01-17', 800.00, 6, 2, 107, 11),
(1009, '2023-01-18', 600.00, 8, 3, 108, 12),
(1010, '2023-01-19', 40.00, 7, 1, 109, 7),
(1011, '2023-01-20', 1.50, 1, 5, 110, 8),
(1012, '2023-01-21', 25.00, 9, 2, 111, 2),
(1013, '2023-01-22', 800.00, 10, 3, 112, 15),
(1014, '2023-01-23', 400.00, 11, 1, 113, 16),
(1015, '2023-01-24', 20.00, 12, 4, 114, 13),
(1016, '2023-01-25', 100.00, 13, 2, 115, 14),
(1017, '2023-01-26', 3.00, 14, 1, 116, 9),
(1018, '2023-01-27', 15.00, 15, 3, 117, 17),
(1019, '2023-01-28', 50.00, 16, 2, 118, 19),
(1020, '2023-01-29', 150.00, 17, 5, 119, 20);
```

SELECT * FROM [Order];

	OrderID	OrderDate	TotalAmount	CustomerID	ShipperID	EmployeeID	ProductID
1	1001	2023-01-10	1200.00	1	1	101	1
2	1002	2023-01-11	25.00	2	2	102	2
3	1003	2023-01-12	45.00	1	1	103	3
4	1004	2023-01-13	300.00	3	3	104	4
5	1005	2023-01-14	150.00	5	2	105	5
6	1006	2023-01-15	1200.00	4	1	101	1
7	1007	2023-01-16	5.00	2	4	106	10
8	1008	2023-01-17	800.00	6	2	107	11
9	1009	2023-01-18	600.00	8	3	108	12
10	1010	2023-01-19	40.00	7	1	109	7
11	1011	2023-01-20	1.50	1	5	110	8
12	1012	2023-01-21	25.00	9	2	111	2
13	1013	2023-01-22	800.00	10	3	112	15
14	1014	2023-01-23	400.00	11	1	113	16
15	1015	2023-01-24	20.00	12	4	114	13
16	1016	2023-01-25	100.00	13	2	115	14
17	1017	2023-01-26	3.00	14	1	116	9
18	1018	2023-01-27	15.00	15	3	117	17
19	1019	2023-01-28	50.00	16	2	118	19
20	1020	2023-01-29	150.00	17	5	119	20

Aggregate Operations:

<u>Finding the most expensive product using **MAX**</u>

SELECT MAX(Price) AS [Max Price] FROM Products;

	Max Price
1	1200.00

Conclusion: The most expensive product is a laptop at 1200.00 price units.

<u>Finding the lowest order amount using **MIN**</u>

SELECT MIN(TotalAmount) AS [Min Order] FROM [Order];

	Min Order
1	1.50

Conclusion: The pen order with orderID 1011 is the cheapest order made.

<u>Finding the average price of all products using **AVG**</u>

SELECT AVG(Price) AS [Average Product Price] FROM Products;

	Average Product Price
1	248.125000

Conclusion: 248.12 is the cost of an average product sold.

Counting orders by customer using **GROUP BY**

```
SELECT CustomerID,
COUNT(OrderID) AS [Order Count]
FROM [Order]
GROUP BY CustomerID;
```

	CustomerID	Order Count
1	1	3
2	2	2
3	3	1
4	4	1
5	5	1
6	6	1
7	7	1
8	8	1
9	9	1
10	10	1
11	11	1
12	12	1
13	13	1
14	14	1
15	15	1
16	16	1
17	17	1

Listing products from most expensive to the cheapest using **ORDER BY**

SELECT * FROM Products ORDER BY Price DESC;

	ProductID	ProductName	Price
1	1	Laptop	1200.00
2	11	Phone	800.00
3	15	Sofa	800.00
4	12	Tablet	600.00
5	16	Table	400.00
6	4	Monitor	300.00
7	6	Desk	250.00
8	5	Chair	150.00
9	20	Watch	150.00
10	14	Headphones	100.00
11	19	Bag	50.00
12	3	Keyboard	45.00
13	7	Lamp	40.00
14	2	Mouse	25.00
15	13	Charger	20.00
16	17	Book	15.00
17	18	Magazine	8.00
18	10	Stapler	5.00
19	9	Notebook	3.00
20	8	Pen	1.50

Join Operations:

Joining all customers (even if they haven't ordered) and their orders using **LEFT JOIN**

SELECT Customer.CustomerName, [Order].OrderID, [Order].OrderDate

FROM Customer

LEFT JOIN [Order] ON Customer.CustomerID = [Order].CustomerID;

	CustomerName	OrderID	OrderDate
1	Alice Smith	1001	2023-01-10
2	Alice Smith	1003	2023-01-12
3	Alice Smith	1011	2023-01-20
4	Bob Jones	1002	2023-01-11
5	Bob Jones	1007	2023-01-16
6	Charlie Brown	1004	2023-01-13
7	Diana Prince	1006	2023-01-15
8	Evan Wright	1005	2023-01-14
9	Fiona Green	1008	2023-01-17
10	George Hall	1010	2023-01-19
11	Hannah Lee	1009	2023-01-18
12	Ian Scott	1012	2023-01-21
13	Jane Doe	1013	2023-01-22
14	Kyle Reese	1014	2023-01-23
15	Laura Croft	1015	2023-01-24
16	Mike Ross	1016	2023-01-25
17	Nancy Drew	1017	2023-01-26
18	Oscar Wilde	1018	2023-01-27
19	Paul Rudd	1019	2023-01-28
20	Quinn Fabray	1020	2023-01-29

<u>Joining only customers who had placed the order with the orders using **INNER JOIN**</u>

SELECT Customer.CustomerName, [Order].OrderID, [Order].TotalAmount

FROM Customer

INNER JOIN [Order] ON Customer.CustomerID = [Order].CustomerID;

	CustomerName	OrderID	TotalAmount
1	Alice Smith	1001	1200.00
2	Bob Jones	1002	25.00
3	Alice Smith	1003	45.00
4	Charlie Brown	1004	300.00
5	Evan Wright	1005	150.00
6	Diana Prince	1006	1200.00
7	Bob Jones	1007	5.00
8	Fiona Green	1008	800.00
9	Hannah Lee	1009	600.00
10	George Hall	1010	40.00
11	Alice Smith	1011	1.50
12	Ian Scott	1012	25.00
13	Jane Doe	1013	800.00
14	Kyle Reese	1014	400.00
15	Laura Croft	1015	20.00
16	Mike Ross	1016	100.00
17	Nancy Drew	1017	3.00
18	Oscar Wilde	1018	15.00
19	Paul Rudd	1019	50.00
20	Quinn Fabray	1020	150.00

Joining all the orders and their matching customers using **RIGHT JOIN**

SELECT Customer.CustomerName, [Order].OrderID, [Order].OrderDate

FROM Customer

RIGHT JOIN [Order] ON Customer.CustomerID = [Order].CustomerID;

	CustomerName	OrderID	OrderDate
1	Alice Smith	1001	2023-01-10
2	Bob Jones	1002	2023-01-11
3	Alice Smith	1003	2023-01-12
4	Charlie Brown	1004	2023-01-13
5	Evan Wright	1005	2023-01-14
6	Diana Prince	1006	2023-01-15
7	Bob Jones	1007	2023-01-16
8	Fiona Green	1008	2023-01-17
9	Hannah Lee	1009	2023-01-18
10	George Hall	1010	2023-01-19
11	Alice Smith	1011	2023-01-20
12	Ian Scott	1012	2023-01-21
13	Jane Doe	1013	2023-01-22
14	Kyle Reese	1014	2023-01-23
15	Laura Croft	1015	2023-01-24
16	Mike Ross	1016	2023-01-25
17	Nancy Drew	1017	2023-01-26
18	Oscar Wilde	1018	2023-01-27
19	Paul Rudd	1019	2023-01-28
20	Quinn Fabray	1020	2023-01-29

Joining both Customer table and Order table by CustomerID using **FULL OUTER JOIN**

SELECT Customer.CustomerName, [Order].OrderID

FROM Customer

FULL OUTER JOIN [Order] ON Customer.CustomerID = [Order].CustomerID;

	CustomerName	OrderID
1	Alice Smith	1001
2	Alice Smith	1003
3	Alice Smith	1011
4	Bob Jones	1002
5	Bob Jones	1007
6	Charlie Brown	1004
7	Diana Prince	1006
8	Evan Wright	1005
9	Fiona Green	1008
10	George Hall	1010
11	Hannah Lee	1009
12	Ian Scott	1012
13	Jane Doe	1013
14	Kyle Reese	1014
15	Laura Croft	1015
16	Mike Ross	1016
17	Nancy Drew	1017
18	Oscar Wilde	1018
19	Paul Rudd	1019
20	Quinn Fabray	1020